

An alternative to the Quantum Fourier Transform

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The hidden subgroup problem (HSP) yielding exponential quantum speedup over the best known classical algorithms, uses the Quantum Fourier Transform (QFT). Standard HSP algorithms require the QFT, whose circuit depth and long-range gates remain challenging for near-term hardware. We propose a departure from conventional QFT approximations: instead of mimicking its gate structure, we identify two functional properties that are exploited by the QFT to enable HSP. Firstly the shift invariance property of the QFT allows for the removal of a random overall shift. Secondly, the information about the hidden subgroup generator needs to remain accessible in the computational basis, and we quantify that via the Fisher information. We construct a family of shallow circuits that preserve shift invariance. Numerical analysis shows these circuits retain Fisher information. We demonstrate successful reconstruction using a neural network and analyze performance under noise, outlining a potential path to practical HSP implementations.

Introduction. — The exponential quantum speedups demonstrated by landmark protocols—such as Shor’s algorithms for prime factorization and discrete logarithms [? ?], as well as Simon’s algorithm [?]—can all be fundamentally abstracted into a single mathematical task: finding an unknown subgroup. This unifying framework is formally known as the hidden subgroup problem (HSP) [1–3]. Standard HSP algorithms require the quantum Fourier transform (QFT), whose $\mathcal{O}(n^2)$ gate complexity remains a critical bottleneck. For a 100-qubit system, the QFT demands thousands of circuit layers, which vastly exceeds current hardware capabilities. For instance, IBM’s 127-qubit Eagle processor is limited to a depth of 60 even with extensive error mitigation [?], and Google’s 105-qubit Willow processor reaches its operational threshold at 40 cycles [?]. Consequently, the complex, long-range controlled-phase interactions necessitated by the standard QFT present significant experimental challenges.

To circumvent the $\mathcal{O}(n^2)$ complexity bottleneck inherent in the standard QFT, existing mitigation strategies have primarily focused on reducing the hardware execution costs while approximating the QFT. These efforts have successfully bounded gate complexity through techniques such as truncated controlled rotations [?] and optimized CNOT routing tailored for restricted hardware topologies [?]. Furthermore, advanced methods like dynamic mid-circuit measurements [?] and quantum circuit cutting [? ?] have been employed to trade coherent quantum depth for classical post-processing resources. While these optimizations have enabled record experimental QFT fidelities [?], they remain fundamentally tethered to the structural replication of the QFT itself. Consequently, such approaches are ultimately limited by the inherent constraints of gate-level simplifications and the exponentially escalating overhead of classical reconstruction, which often becomes computationally prohibitive as the system size scales.

In this Letter, we propose a fundamental paradigm shift: seeking a functional alternative to the QFT rather

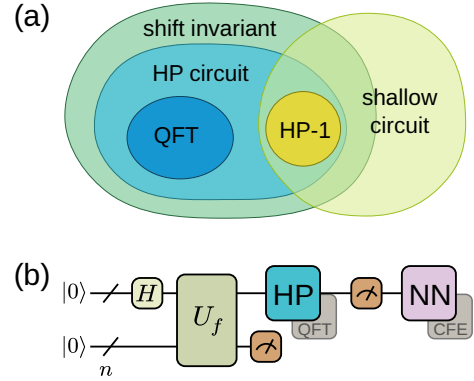


FIG. 1. **Shallow shift-invariant alternative to the QFT.** (a) Shift invariance is a critical property of the QFT. We identify a circuit family, denoted as HP- k , that inherently preserves the property. Within this broader family, HP-1 specifically serves as shallow-depth gates that are friendly to experiment. (b) In the proposed HSP algorithmic routine, the deep QFT block (the left gray gate) is directly replaced by the HPcircuit to extract subgroup information prior to measurement. A neural network then decodes the measurement outcomes, bypassing the standard continued fraction expansion (the right gray gate) typically used in classical post-processing.

than a mere structural approximation. Our fundamental motivation is directly illustrated in Fig. 1(a). The standard QFT extracts hidden subgroup information primarily through shift invariance, which ensuring that measurement statistics remain independent of random coset representatives, but it inherently requires deep circuits that are prohibitive for near-term hardware. To circumvent this bottleneck, we abandon structural emulation and instead target the crucial intersection between the broader space of shift-invariant transformations and the space of shallow circuits. We construct a family of hardware-aware quantum circuits, designated as HPcircuits, that natively preserves this shift invariance. Within this broader family, we highlight HP-1, which successfully occupies this exact intersection by serving

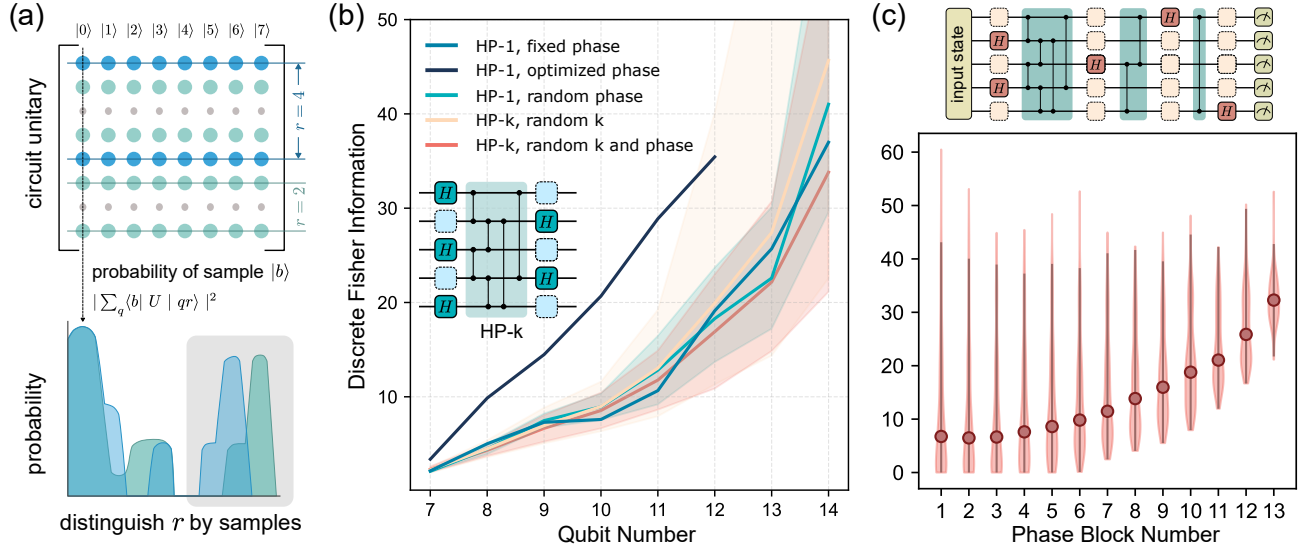


FIG. 2. Numerical Fisher-information benchmarks for the candidate circuit families. (a) Minimum discrete Fisher information as a function of the number of qubits for the QFT reference and the HP-family circuits. The HP-family curves remain below the QFT benchmark, but all increase with system size. (b) Fisher-information distributions as a function of the number of layers, showing how additional layers increase the retained information. Appendix E shows that, over the sampled qubit window, both the QFT reference and the HP-family curves are well described by the same log-linear model $\ln I_{\min}(n) = kn + b$, corresponding to exponential growth in n , with smaller effective exponents for the HP-family circuits.

as shallow-depth gates friendly to experiment. By combining this depth advantage with the retention of Fisher information, HP-1 serves as a robust functional substitute, offering a highly practical and experimentally feasible pathway for HSP implementation over finite abelian groups on near-term hardware.

Structurally, these circuits interleave transversal Hadamard gates with parameterized phase blocks. This architecture is natively compatible with near-term hardware, as sequences of phase gates are significantly more easily achieved across multiple experimental platforms, including linear optics [?], neutral-atom systems [? ?], ion trap systems [? ?], and NMR quantum computing [? ?]. Alongside shift invariance, the viability of our approach relies on the retention of Fisher information, which quantifies the extractable information regarding the hidden subgroup generator r . Numerical scaling analysis demonstrates that shallow HPCircuits retain substantial Fisher information. Despite a smaller scaling exponent k relative to the full QFT, they maintain an equivalent exponential growth trend $FI_n \sim e^{kn}$, thereby enabling reliable subgroup reconstruction via efficient neural-network-based decoding.

The proposed quantum-classical protocol is illustrated in Fig. 1(b). We directly replace the deep QFT block with the shallow HPCircuit in the original protocol for solving HSP over finite abelian group. A neural network then decodes the measurement outcomes, substituting the standard continued fraction expansion typically used in classical post-processing. To theoretically guarantee the scalability of this approach, we validate the

growth model of the Fisher information with the various of qubit number, confirming that the proposed circuit family systematically retains sufficient extractable information at scale. Furthermore, we numerically benchmark this hybrid scheme on Shor’s algorithm; the results demonstrate that the new circuit, combined with neural post-processing, can successfully recover the period and achieve factorization. Finally, to ensure practical viability, we analyze the robustness of the proposed circuits under a global depolarizing noise model.

HSP and Shift Invariance. — The HSP provides a unifying mathematical framework for exponential quantum speedups, abstracting landmark protocols into a single task: identifying an unknown subgroup $H \leq G$ using an oracle function that is constant on the cosets of H and distinct on different cosets [? ? ?]. For finite abelian groups, there is a standard quantum routine that solves this problem efficiently, where we put more details about HSP and its solver in Appendix A. The standard quantum routine is illustrated in Fig. 1(b) with gray blocks.

Crucially, the QFT extracts the hidden subgroup information primarily through shift invariance which is a critical property ensuring that the output measurement statistics remain strictly independent of random coset representatives. We formulate the shift invariance as following:

Definition 1 (Shift invariance). For $g \in G$, define the shift operator by $T_g |h\rangle = |h + g\rangle$. For a subgroup $H \leq G$,

define the normalized subgroup state

$$|H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |h\rangle. \quad (1)$$

A unitary U on the group register is shift-invariant for HSP sampling if

$$|\langle x|U|H\rangle|^2 = |\langle x|UT_g|H\rangle|^2, \quad \forall x, g \in G, \quad \forall H \leq G. \quad (2)$$

This review makes clear that the key operational requirement is the coherent, information-preserving interference of different hidden-subgroup cosets. From this perspective, shift invariance is not merely a desirable feature, but a necessary condition for any circuit intended to realize a Fourier-type transformation for the HSP. This condition then serves to narrow the class of admissible circuit constructions.

Based on this, we then derive the necessary conditions that any candidate circuit must satisfy.

Theorem 1 (Necessary condition of shift invariance). *Suppose U is a unitary on the group register that satisfies the shift-invariance condition above. Then, for every $x, y \in G$,*

$$|\langle x|U|y\rangle| = \frac{1}{\sqrt{|G|}}. \quad (3)$$

Equivalently, there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (4)$$

The proof is provided in Appendix B.

Building on this conceptual framework, we introduce a formal definition of a shift-invariant circuit, showing how this formulation captures the essential operational requirements of the HSP measurement stage, substantially narrows the space of viable alternatives to the quantum Fourier transform (QFT), and clarifies the structural properties required for successful circuit construction.

HP: A shift invariant family of circuits. — (Old) To bridge the gap between the necessary conditions for shift invariance and explicit circuit construction, we introduce a parameterized family of quantum circuits, denoted as the HP-k circuits.

(New) Guided by the necessary conditions for shift invariance, we construct a parameterized family of quantum circuits, denoted as the HP-k circuits, which explicitly realizes these requirements.

The fundamental design principle of this architecture is to systematically interleave transversal layers of single-qubit Hadamard gates with parameterized controlled-phase operations. By doing so, the circuit hierarchically builds the precise phase relationships required for subgroup interference.

Let H_i denote the Hadamard gate applied to qubit i . We partition the circuit into L sequential stages. For a given stage l , we define a subset of control qubits Λ_l and a corresponding subset $\bar{\Lambda}_l$ consisting of the remaining qubits that have not yet been used as targets. These subsets are coupled via a controlled-phase block, denoted as $\text{CP}_{\Lambda_l, \bar{\Lambda}_l}(\theta_{\Lambda_l \bar{\Lambda}_l})$, which applies pairwise operations $\text{CP}_{jk}(\theta_{jk})$ between each qubit $j \in \bar{\Lambda}_l$ and $k \in \Lambda_l$. Building upon these notations, we formally define the circuit architecture:

Definition 2 (HP-k circuit). *A quantum circuit is classified as an HP-k circuit if it is constructed using the following architecture:*

$$U_{\text{HP}}^{(L)}(\theta) = \bigotimes_{i \in \Lambda_L} H_i \prod_{l=L-1}^1 \left(\text{CP}_{\Lambda_l, \bar{\Lambda}_l}(\theta_{\Lambda_l \bar{\Lambda}_l}) \bigotimes_{i \in \Lambda_l} H_i \right), \quad (5)$$

where the controlled-phase block is given by:

$$\text{CP}_{\Lambda_l, \bar{\Lambda}_l}(\theta_{\Lambda_l \bar{\Lambda}_l}) = \prod_{j \in \bar{\Lambda}_l} \prod_{k \in \Lambda_l} \text{CP}_{jk}(\theta_{jk}). \quad (6)$$

By making this architecture explicit, we identify a concrete family of circuits that strictly satisfies the previously derived shift-invariance conditions, thereby providing a nontrivial alternative for the standard QFT circuit. Importantly, this proposed family encompasses a strictly broader set of transformations than those generated by conventional QFT constructions. This generalization demonstrates that shift invariance is not a unique feature of the QFT, but rather a robust property that can naturally arise within a significantly larger design space of practically useful quantum circuits.

Theorem 2. *HP-k circuits are shift invariant under group $\prod_i \mathbb{Z}_{2^{m_i}}$.*

HP-0 for hidden subgroup on $\mathbb{Z}_{2^n}$. — We first consider the cyclic group $\mathbb{Z}_{2^n}$, which is naturally represented by an n -qubit computational basis and therefore provides a clean setting for isolating the effect of the circuit transformation. In this case, the subgroup structure is particularly simple: each subgroup is generated by a power of two, and the corresponding coset states have a regular binary structure. For example, if $H = \{q2^p : q = 0, \dots, 2^{n-p} - 1\}$, then, up to normalization and bit-ordering convention, applying a transversal Hadamard layer to a coset state gives

$$H^{\otimes n} \sum_{h \in H} |c+h\rangle \propto \sum_{k \in \{0,1\}^p} (-1)^{k \cdot c} |0\rangle^{\otimes(n-p)} |k\rangle. \quad (7)$$

Thus the dependence on the coset representative c appears only through phases, while the support of the measurement distribution reveals the number of fixed zero

bits and hence the hidden subgroup parameter. The detailed calculation, together with the corresponding sampling algorithm using transversal Hadamard gates, is provided in Appendix C. A transversal layer of Hadamard gates, which we identify as the HP-0 circuit, is therefore already sufficient to remove the irrelevant coset shift at the level of measurement probabilities and to retain information about the hidden subgroup in the idealized finite-group setting. Thus, HP-0 provides a useful baseline demonstrating that a shallow shift-invariant transformation can be sufficient for a meaningful subclass of HSP instances.

At the same time, the $\mathbb{Z}_{2^n}$ case should be regarded as an instructive but restricted regime. Its subgroup structure is highly constrained, and several such instances admit efficient classical treatment, so this setting alone does not capture the full algorithmic role played by the QFT. The more demanding regime arises when one uses a finite-dimensional register to approximate a problem over $\mathbb{Z}$ and to recover a subgroup such as $\mathbb{Z}_q$. In that situation, the sampled state need not be an exact subgroup state of the truncated register, and the robustness of the standard QFT becomes essential. This motivates the extension from HP-0 to deeper members of the HPfamily, which are designed to preserve shift-invariant sampling structure while allowing greater expressive power than a single Hadamard layer.

Retention of Fisher information on hidden subgroup.—

To justify the post-processing approach more systematically, we ask whether the hidden subgroup generator can be estimated using only a polynomial number of measurement samples. In our setting, this generator plays the role of the parameter in the output distribution, so the relevant question is how sensitive the measured probabilities are to changes in that generator. Fisher information provides a natural metric for this purpose, since it quantifies the amount of recoverable information about the generator contained in the samples. As a benchmark, for the standard QFT we obtain

$$I_{\text{QFT}}(r) = \frac{8}{\pi R} \sum_{k=1}^{R-1} (R-k) \left[\frac{k\pi^3}{3} + \sum_{j=1}^{k-1} \frac{k-j}{j^3 r^3} \Gamma_j(r) \right],$$

$$\Gamma_j(r) = 2\pi j r \cos(2\pi j r) + (2\pi^2 j^2 r^2 - 1) \sin(2\pi j r), \quad (8)$$

which in the large- r regime reduces to $I_{\text{QFT}}(r) \approx \frac{4\pi^2}{9}(R^2 - 1)$. The detailed derivation is given in Appendix D.

HP-1 retains Fisher information with fewer layers than QFT. — We compute the Fisher information numerically as a function of both circuit depth and system size. The observed scaling offers strong evidence that the proposed circuit family preserves enough information for practical inference. Most notably, HP-1 retains a level of Fisher information comparable to that of the full QFT

while remaining considerably shallower, thereby reducing the depth requirements of the transformation stage. The comparison is sharpened in Appendix E: although the HP-family Fisher information is systematically smaller than the QFT benchmark, the numerical data are well fit by the same log-linear ansatz $\ln I_{\min}(n) = kn + b$, so over the sampled range both exhibit the same exponential-in- n growth model, differing primarily in the fitted exponent.

Simulation of HSP with HP-1 and efficient neural net post-processing. — We use a neural network as a learn-

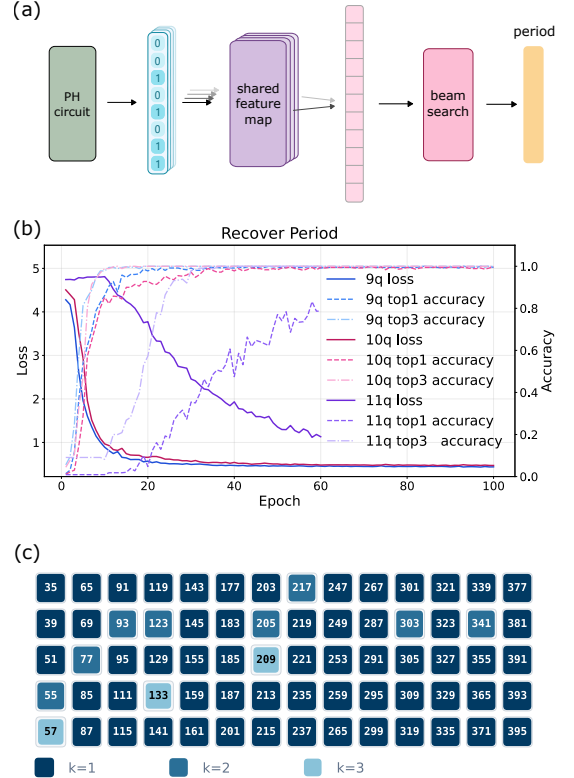


FIG. 3. Classical post-processing pipeline for subgroup recovery from HP-1 measurement data. The measured bit strings produced after the shallow quantum circuit are passed through a shared feature map and then ranked by a beam-search decoder to infer the candidate period. The figure illustrates that the measurement outcomes still retain period-dependent structure after the HP transformation, which can be extracted by an efficient neural post-processing stage rather than by an explicit QFT inversion.

able classical post-processing layer that takes measurement outcomes from the HP-1 circuit and predicts the hidden subgroup parameter. This experiment is intended to test the practical recoverability of subgroup information, rather than only reporting information-theoretic quantities in the abstract. The resulting performance shows that the measurement distributions produced by these shallow circuits still contain highly usable structure, and that an efficient classical decoder can success-

fully exploit it.

Robustness under noise.— We also evaluate the previously trained HP-1 decoder under a post-circuit global depolarizing channel. The held-out recovery accuracy remains essentially unchanged up to noise strength $\eta \approx 0.215$, while stronger noise causes a marked drop: at $\eta = 0.464$ the accuracy is 0.491 for 9 qubits and 0.881 for 10 qubits, and at the fully depolarizing point $\eta = 1$ it falls to chance. Detailed settings, complete numerical results, and the corresponding interpretation are given in Appendix F.

Summary and outlook. — In summary, we have established a framework for solving the Hidden Subgroup Problem using shift-invariant circuits that circumvent the depth and gate-connectivity bottlenecks of the traditional QFT. The main remaining limitation of the current proposal is its generalization ability. Extending this approach, together with the corresponding classical post-

processing, to more complex group structures requires further investigation and will form the basis of our future work.

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APPENDIX

A. Hidden Subgroup Problem and Fourier Sampling

In this section, we outline the standard method for solving the Hidden Subgroup Problem (HSP) over finite abelian groups [1–3]. Formally, the HSP is defined as follows:

Definition 3 (Hidden Subgroup Problem). *Given a function $f : G \rightarrow X$ defined on a group G , with the promise that there exists a hidden subgroup $H \leq G$ such that for all $g_1, g_2 \in G$*

$$f(g_1) = f(g_2) \iff g_1 - g_2 \in H \quad (\text{A1})$$

The objective of this task is to identify the hidden subgroup H .

In the context of this paper, we restrict our attention to finite abelian groups. For such groups, the fundamental theorem of finite abelian groups allows us to express the group structure as a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p_1} \times \mathbb{Z}_{p_2} \times \cdots \times \mathbb{Z}_{p_k}. \quad (\text{A2})$$

The standard quantum procedure for solving the hidden subgroup problem begins by preparing a uniform superposition over all elements of the group G [1, 4]. We initialize the system in the state

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |0\rangle, \quad (\text{A3})$$

where $|G|$ is the element num of the group G . The first register holds the group elements and the second register is initialized to a standard basis state. Next, we evaluate the quantum oracle corresponding to the function f , which acts on the two registers and entangles the group elements with their corresponding function values

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |f(g)\rangle. \quad (\text{A4})$$

Since the function f is constant on the cosets of the hidden subgroup H and distinct on different cosets, it is highly advantageous to group the terms in the superposition according to these cosets.

Before proceeding, it is useful to formalize the state notation for a subset. For any subset $S \subseteq G$, we define its corresponding normalized quantum state $|S\rangle$ as the uniform superposition over all elements within the set:

$$|S\rangle = \frac{1}{\sqrt{|S|}} \sum_{s \in S} |s\rangle. \quad (\text{A5})$$

Using this definition, the normalized state for a specific coset $c + H$ is directly given by

$$|c + H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |c + h\rangle. \quad (\text{A6})$$

By applying this coset state definition, the post-oracle state can be elegantly rewritten as a superposition of these orthogonal coset states, each perfectly entangled with its unique function evaluation:

$$|\psi\rangle = \sqrt{\frac{|H|}{|G|}} \sum_{c \in C} |c + H\rangle |f(c)\rangle, \quad (\text{A7})$$

where C is the set of all possible coset representatives. For simplicity, we assume that the second register is measured immediately after the oracle query. Measuring the second register yields a specific outcome $f(c)$ with a uniform probability of $|H|/|G|$. Consequently, the first register collapses into the corresponding pure coset state:

$$|\psi_{\text{post}}\rangle = |c + H\rangle. \quad (\text{A8})$$

From this point forward, our analysis will focus exclusively on the quantum state of this first register.

These two descriptions are different at the state level, but they induce the same Fourier-basis statistics because every coset state has the same measurement distribution after the quantum Fourier transform.

A direct measurement in the computational basis is uninformative. Indeed, using the definition of the coset state,

$$\langle g | c + H \rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} \langle g | c + h \rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} \delta_{g, c+h}. \quad (\text{A9})$$

For fixed c , the map $h \mapsto c + h$ is a bijection from H to the coset $c + H$. Therefore the sum $\sum_{h \in H} \delta_{g, c+h}$ equals 1 when $g \in c + H$ and equals 0 when $g \notin c + H$. Hence

$$\Pr[g | c, H] = |\langle g | c + H \rangle|^2 = \begin{cases} 1/|H|, & g \in c + H, \\ 0, & g \notin c + H, \end{cases} \quad (\text{A10})$$

so the observed element is uniform on the unknown coset. Averaging over uniformly random cosets gives the marginal distribution

$$\Pr[g] = \frac{1}{|G|}, \quad (\text{A11})$$

which contains no direct information about the hidden subgroup H .

To extract meaningful information, we must apply the quantum Fourier transform (QFT) over the group G before measuring [1, 2]. Let $\chi_y(g)$ denote the group characters. The QFT maps a computational basis state $|g\rangle$ as follows:

$$\text{QFT} |g\rangle = \frac{1}{\sqrt{|G|}} \sum_{y \in G} \chi_y(g) |y\rangle. \quad (\text{A12})$$

Applying this transformation to our pure coset state $|c + H\rangle$ yields:

$$\begin{aligned} \text{QFT} |c + H\rangle &= \frac{1}{\sqrt{|H|}} \sum_{h \in H} \text{QFT} |c + h\rangle \\ &= \frac{1}{\sqrt{|H||G|}} \sum_{h \in H} \sum_{y \in G} \chi_y(c + h) |y\rangle \\ &= \frac{1}{\sqrt{|H||G|}} \sum_{y \in G} \chi_y(c) \left(\sum_{h \in H} \chi_y(h) \right) |y\rangle. \end{aligned} \quad (\text{A13})$$

By the orthogonality of group characters, the sum $\sum_{h \in H} \chi_y(h)$ evaluates to $|H|$ if the character χ_y is trivial on H (i.e., y belongs to the orthogonal subgroup $H^\perp$), and evaluates to 0 otherwise. Thus, the state simplifies to a superposition over $H^\perp$:

$$\text{QFT } |c + H\rangle = \sqrt{\frac{|H|}{|G|}} \sum_{y \in H^\perp} \chi_y(c) |y\rangle. \quad (\text{A14})$$

Measuring this state in the computational basis gives the probability of observing an outcome $y \in H^\perp$:

$$\Pr[y] = \left| \sqrt{\frac{|H|}{|G|}} \chi_y(c) \right|^2 = \frac{|H|}{|G|} |\chi_y(c)|^2 = \frac{|H|}{|G|}. \quad (\text{A15})$$

Crucially, because $|\chi_y(c)|^2 = 1$ for all group characters, the resulting measurement distribution is uniformly distributed over $H^\perp$ and is strictly independent of the random coset representative c . This shift invariance is the fundamental property that allows the extraction of subgroup information by effectively removing the unknown shift at the probability level.

The quantum Fourier transform over G is defined using the character group $\hat{G}$ by

$$F_G |g\rangle = \frac{1}{\sqrt{|G|}} \sum_{\chi \in \hat{G}} \chi(g) |\chi\rangle, \quad (\text{A16})$$

where $\{|\chi\rangle : \chi \in \hat{G}\}$ is the Fourier basis and each character $\chi : G \rightarrow \mathbb{C}^\times$ is a group homomorphism. Here $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ denotes the multiplicative group of nonzero complex numbers, so with additive notation on G one has $\chi(g_1 + g_2) = \chi(g_1)\chi(g_2)$. For finite abelian G , every character has unit modulus. Applying F_G to a coset state gives

$$\begin{aligned} F_G |c + H\rangle &= \frac{1}{\sqrt{|H|}} \sum_{h \in H} F_G |c + h\rangle \\ &= \frac{1}{\sqrt{|H||G|}} \sum_{\chi \in \hat{G}} \sum_{h \in H} \chi(c + h) |\chi\rangle \\ &= \frac{1}{\sqrt{|H||G|}} \sum_{\chi \in \hat{G}} \chi(c) \left(\sum_{h \in H} \chi(h) \right) |\chi\rangle. \end{aligned} \quad (\text{A17})$$

The character sum over H satisfies

$$\sum_{h \in H} \chi(h) = \begin{cases} |H|, & \chi \in H^\perp, \\ 0, & \chi \notin H^\perp, \end{cases} \quad H^\perp = \{\chi \in \hat{G} : \chi(h) = 1 \text{ for all } h \in H\}. \quad (\text{A18})$$

Hence

$$F_G |c + H\rangle = \sqrt{\frac{|H|}{|G|}} \sum_{\chi \in H^\perp} \chi(c) |\chi\rangle. \quad (\text{A19})$$

The Fourier-basis measurement distribution is therefore

$$p(\chi | c, H) = \begin{cases} |H|/|G|, & \chi \in H^\perp, \\ 0, & \chi \notin H^\perp, \end{cases} \quad (\text{A20})$$

which is independent of the coset representative c . The same formula holds for the mixed state ρ_H , since it is a convex combination of pure coset states with identical Fourier statistics.

Repeating the procedure therefore produces samples from $H^\perp$. In the finite abelian setting, classical post-processing reconstructs $H^\perp$ from the sample span and then recovers the hidden subgroup using the duality identity [1, 3, 5]

$$(H^\perp)^\perp = H. \quad (\text{A21})$$

Fourier sampling thus converts hidden translational structure in the oracle into explicit linear constraints in the dual group.

B. Necessary condition of shift invariance

Here we make explicit the probability-level shift-invariance condition used in Fourier sampling. Let G be a finite abelian group with $N = |G|$, and let $\{|g\rangle : g \in G\}$ denote the computational basis. For $g \in G$, define the shift operator

$$T_g |h\rangle = |h + g\rangle. \quad (\text{B1})$$

For a subgroup $H \leq G$, write the normalized subgroup state as

$$|H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |h\rangle. \quad (\text{B2})$$

The shift-invariance requirement relevant to HSP sampling is that the measurement distribution after applying U is independent of the coset representative:

$$|\langle x|U|H\rangle|^2 = |\langle x|UT_g|H\rangle|^2, \quad \forall x, g \in G, \forall H \leq G. \quad (\text{B3})$$

Thus $U|H\rangle$ and $U|g+H\rangle$ may differ by phases, but they must produce the same probabilities in the computational basis.

This is the operational notion of shift invariance relevant for HSP sampling. The random coset representative should disappear from the observed distribution after the transformation, even though it need not disappear from the output state itself. Any candidate replacement for the QFT must therefore satisfy Eq. (B3) if it is to reproduce the same basic sampling mechanism.

The next question is how restrictive Eq. (B3) already is at the level of matrix elements. A useful simplification is that one can extract a necessary condition from the trivial subgroup $H = \{0\}$ alone. In that case the subgroup state is just a computational-basis state, so the shift-invariance requirement becomes a direct constraint on how the columns of U are seen from each measurement outcome x .

The idea is simple. If the output probabilities for $|0\rangle$ and $|g\rangle$ must agree for every shift g , then for each fixed x all entries in the x -th row of U must have the same modulus. Unitarity then forces that common modulus to be exactly $1/\sqrt{|G|}$. This is the content of Theorem 1, which we restate here in the precise form used in this appendix.

Theorem (Necessary modulus condition). *Suppose U is a unitary on the group register that satisfies Eq. (B3). Then, for every $x, y \in G$,*

$$|\langle x|U|y\rangle| = \frac{1}{\sqrt{|G|}}. \quad (\text{B4})$$

Equivalently, there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (\text{B5})$$

Proof of Theorem 1. Take the trivial subgroup $H = \{0\}$. Then $|H\rangle = |0\rangle$ and $T_g|H\rangle = |g\rangle$ for every $g \in G$. Equation (B3) therefore implies

$$|\langle x|U|0\rangle|^2 = |\langle x|U|g\rangle|^2, \quad \forall x, g \in G. \quad (\text{B6})$$

With the convention $U_{xy} = \langle x|U|y\rangle$, this says that for each fixed output label x , all entries in the x -th row of U have the same squared modulus. Denote this common value by a_x . Since U is unitary, each row is normalized:

$$1 = \sum_{y \in G} |U_{xy}|^2 = N a_x. \quad (\text{B7})$$

Hence $a_x = 1/N$ for every $x \in G$. Therefore every matrix element satisfies

$$|U_{xy}| = \frac{1}{\sqrt{N}} = \frac{1}{\sqrt{|G|}}. \quad (\text{B8})$$

Because this modulus is nonzero, each entry admits a polar decomposition

$$U_{xy} = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}, \quad \theta_{xy} \in \mathbb{R}. \quad (\text{B9})$$

This proves the claim. $\square$

C. Transversal Hadamard sampling and efficiency guarantee

We now give the calculation underlying the HP-0 discussion in the main text. Consider $G = \mathbb{Z}_{2^n}$ represented by n computational-basis qubits, and use the bit ordering in which a multiple of 2^p has the form

$$h = h_1 h_2 \cdots h_{n-p} 0 \cdots 0. \quad (\text{C1})$$

The subgroup generated by 2^p is

$$H_p = \{q2^p : q = 0, \dots, 2^{n-p} - 1\}, \quad (\text{C2})$$

so $|H_p| = 2^{n-p}$. Since adding an element of H_p can vary the first $n-p$ bits, each coset has a representative of the form

$$c = 0 \cdots 0 c_{n-p+1} \cdots c_n. \quad (\text{C3})$$

Thus a normalized coset state can be written as

$$|c + H_p\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} |h_1 \cdots h_{n-p} c_{n-p+1} \cdots c_n\rangle. \quad (\text{C4})$$

For an n -bit string x , the transversal Hadamard transform satisfies

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{k \cdot x} |k\rangle. \quad (\text{C5})$$

Applying this to Eq. (C4) gives

$$\begin{aligned} H^{\otimes n} |c + H_p\rangle &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} (-1)^{\sum_{r=1}^{n-p} k_r h_r + \sum_{r=n-p+1}^n k_r c_r} |k\rangle \\ &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{\sum_{r=n-p+1}^n k_r c_r} \prod_{r=1}^{n-p} \left(\sum_{h_r \in \{0,1\}} (-1)^{k_r h_r} \right) |k\rangle. \end{aligned} \quad (\text{C6})$$

The inner product factor is zero unless $k_1 = \cdots = k_{n-p} = 0$, in which case it equals 2^{n-p} . Therefore

$$H^{\otimes n} |c + H_p\rangle = \frac{1}{\sqrt{2^p}} \sum_{k_{n-p+1}, \dots, k_n \in \{0,1\}} (-1)^{\sum_{r=n-p+1}^n k_r c_r} |0\rangle^{\otimes (n-p)} |k_{n-p+1} \cdots k_n\rangle. \quad (\text{C7})$$

The coset representative c only contributes the phase factor in Eq. (C7). Consequently, the measurement distribution is independent of c : the first $n-p$ bits are always zero, while the remaining p bits are uniformly random.

This immediately gives a simple reconstruction procedure for the idealized $\mathbb{Z}_{2^n}$ HSP. Each application of the oracle-and-measurement routine prepares a random coset state, and Eq. (C7) shows that a subsequent transversal Hadamard layer converts that coset state into a sample whose first $n-p$ bits are deterministically zero while the remaining p bits are unbiased. The resulting sampling-and-recovery routine is summarized in Algorithm 1.

Algorithm 1: Transversal Hadamard Sampling for $H_p \leq \mathbb{Z}_{2^n}$

Input: Oracle access to an HSP instance over $\mathbb{Z}_{2^n}$, sample count m

Output: Hidden subgroup $H_p = \langle 2^p \rangle$

```

1 Initialize an empty list of samples  $\mathcal{B}$ ;
2 for  $i = 1$  to  $m$  do
3   Prepare the uniform superposition  $\frac{1}{\sqrt{2^n}} \sum_{g \in \mathbb{Z}_{2^n}} |g\rangle |0\rangle$ ;
4   Query the oracle and measure the second register to obtain a coset state  $|c_i + H_p\rangle$ ;
5   Apply  $H^{\otimes n}$  to the first register;
6   Measure the first register and record the bit string  $b^{(i)} \in \{0, 1\}^n$ ;
7   Append  $b^{(i)}$  to  $\mathcal{B}$ ;
8 end
9 for  $j = 1$  to  $n$  do
10  Mark bit position  $j$  as fixed if  $b_j^{(i)} = 0$  for all recorded samples;
11  Otherwise mark bit position  $j$  as free;
12 end
13 Let  $p$  be the number of free trailing bit positions;
14 return  $H_p = \langle 2^p \rangle$ ;

```

Algorithm 1 is correct because, under the bit-ordering convention above, the free positions occur exactly in the final p bits. Equivalently, the measurement data have the form

$$0 \cdots 0 * \cdots *, \quad (\text{C8})$$

where each star denotes an unbiased bit. Identifying which positions ever take the value 1 therefore determines p , and hence determines the subgroup $H_p = \langle 2^p \rangle$.

The sample complexity is logarithmic in the number of qubits. For Algorithm 1, fixed bits are never misclassified, because they are exactly zero in every sample. For a free bit, the probability that it appears as zero in all m independent samples is 2^{-m} . By the union bound, the probability of misclassifying at least one free bit is at most

$$p2^{-m} \leq n2^{-m}. \quad (\text{C9})$$

Thus $m \geq \lceil \log_2(n/\epsilon) \rceil$ samples are sufficient for Algorithm 1 to identify the fixed and free bit positions, and therefore the hidden subgroup H_p , with probability at least $1 - \epsilon$.

D. QFT Fisher information for subgroup-generator estimation

To benchmark the proposed circuit family, we treat the hidden subgroup generator r as the parameter to be inferred from the measurement distribution produced by the standard QFT. Consider the truncated periodic state

$$|\psi_r\rangle = \frac{1}{\sqrt{R}} \sum_{q=0}^{R-1} |qr\rangle, \quad (\text{D1})$$

where R is the number of multiples of r supported in the 2^n -dimensional register. For the 2^n -point QFT,

$$|j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i j k / 2^n} |k\rangle, \quad (\text{D2})$$

and therefore

$$|\psi_r\rangle \mapsto \frac{1}{\sqrt{R2^n}} \sum_{k=0}^{2^n-1} \left(\sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right) |k\rangle. \quad (\text{D3})$$

The corresponding measurement probability is

$$p(k|r) = \frac{1}{R2^n} \left| \sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right|^2 = \frac{1}{R2^n} \left(\frac{\sin(\pi R r k / 2^n)}{\sin(\pi r k / 2^n)} \right)^2. \quad (\text{D4})$$

In the regime $r \ll 2^n$, it is convenient to introduce the continuum variable $x = \pi k/2^n \in [0, \pi]$ and approximate the distribution by the density

$$f(x|r) = \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2. \quad (\text{D5})$$

The QFT Fisher information with respect to the generator r is then

$$I_{\text{QFT}}(r) = \int_0^\pi f(x|r) \left[\frac{\partial}{\partial r} \ln f(x|r) \right]^2 dx, \quad (\text{D6})$$

with score function

$$\frac{\partial}{\partial r} \ln f(x|r) = 2Rx \cot(Rrx) - 2x \cot(rx). \quad (\text{D7})$$

The integral representation above makes the statistical meaning of the benchmark transparent, but it still hides the dependence on the support size R inside a strongly oscillatory quotient of sines. For the comparison carried out in the main text, what we need is a formula in which the explicit R -scaling is separated from the oscillatory corrections.

There is one additional simplification that closes this gap. If we write

$$S(u) = \frac{\sin(Ru)}{\sin u}, \quad (\text{D8})$$

then $f(x|r) = S(rx)^2/(\pi R)$, and the score function implies

$$f(x|r) \left[\frac{\partial}{\partial r} \ln f(x|r) \right]^2 = \frac{4x^2}{\pi R} [S'(rx)]^2. \quad (\text{D9})$$

Thus the Fisher-information integral is reduced to the square of a finite trigonometric polynomial. Once that polynomial is expanded into cosine modes, the remaining calculation involves only the moments of x^2 against those modes.

Theorem 3 (QFT Fisher-information benchmark). *For the continuum density*

$$f(x|r) = \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2, \quad x \in [0, \pi], \quad (\text{D10})$$

the Fisher information with respect to the generator r is

$$I_{\text{QFT}}(r) = \frac{4\pi^2}{9}(R^2 - 1) + \frac{8}{3\pi R} \sum_{j=1}^{R-1} (R-j)(R^2 - 2Rj - 2j^2 - 1) M_j(r),$$

$$M_j(r) = \frac{4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2) \sin(2\pi jr)}{(2jr)^3}. \quad (\text{D11})$$

In particular, since $M_j(r) = O(r^{-2})$ for fixed j , the large- r regime is governed by the non-oscillatory term

$$I_{\text{QFT}}(r) \approx \frac{4\pi^2}{9}(R^2 - 1). \quad (\text{D12})$$

To derive Theorem 3, it is convenient to pass from the Fejer-kernel representation of $S(u)^2$ to an exponential representation of $S(u)$ itself. Differentiating that exponential sum yields a finite trigonometric polynomial, and squaring it produces an exact cosine expansion for $[S'(u)]^2$. After that step, the Fisher-information integral reduces to $\int_0^\pi x^2 dx$ and $\int_0^\pi x^2 \cos(2jrx) dx$. The asymptotic formula follows because the oscillatory moments decay as r^{-2} .

Proof. From

$$S(u) = \frac{\sin(Ru)}{\sin u} = \sum_{m=0}^{R-1} e^{i(R-1-2m)u}, \quad (\text{D13})$$

we obtain

$$S'(u) = i \sum_{m=0}^{R-1} (R-1-2m) e^{i(R-1-2m)u}. \quad (\text{D14})$$

Squaring this expression and grouping equal frequencies gives an even cosine series

$$[S'(u)]^2 = A_0 + 2 \sum_{j=1}^{R-1} A_j \cos(2ju), \quad (\text{D15})$$

where

$$A_j = - \sum_{m=0}^{R-1-j} (R-1-2m)(2j-R+1+2m) = \frac{(R-j)(R^2-2Rj-2j^2-1)}{3}, \quad j = 0, \dots, R-1. \quad (\text{D16})$$

In particular,

$$A_0 = \frac{R(R^2-1)}{3}. \quad (\text{D17})$$

Since

$$I_{\text{QFT}}(r) = \frac{4}{\pi R} \int_0^\pi x^2 [S'(rx)]^2 dx, \quad (\text{D18})$$

substituting the cosine expansion yields

$$I_{\text{QFT}}(r) = \frac{4}{\pi R} \left[A_0 \int_0^\pi x^2 dx + 2 \sum_{j=1}^{R-1} A_j \int_0^\pi x^2 \cos(2jrx) dx \right]. \quad (\text{D19})$$

The required moments are

$$\int_0^\pi x^2 dx = \frac{\pi^3}{3} \quad (\text{D20})$$

and

$$\int_0^\pi x^2 \cos(2jrx) dx = \frac{4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2) \sin(2\pi jr)}{(2jr)^3}. \quad (\text{D21})$$

Substituting these expressions gives

$$\begin{aligned} I_{\text{QFT}}(r) &= \frac{4}{\pi R} \left[\frac{R(R^2-1)}{3} \cdot \frac{\pi^3}{3} + \frac{2}{3} \sum_{j=1}^{R-1} (R-j)(R^2-2Rj-2j^2-1) M_j(r) \right] \\ &= \frac{4\pi^2}{9} (R^2-1) + \frac{8}{3\pi R} \sum_{j=1}^{R-1} (R-j)(R^2-2Rj-2j^2-1) M_j(r), \end{aligned} \quad (\text{D22})$$

which is the stated exact formula.

Finally, the oscillatory correction term is subleading because $M_j(r) = O(r^{-2})$ for fixed j . Keeping only the constant contribution therefore gives

$$I_{\text{QFT}}(r) \approx \frac{4\pi^2}{9} (R^2-1). \quad (\text{D23})$$

This proves the theorem. □

Theorem 3 provides the analytical QFT benchmark used in the main text.

E. Log-linear growth fit for the numerical Fisher-information scan

The fixed- r QFT benchmark derived above clarifies the dominant scaling mechanism, but Fig. 2(b) probes a slightly different quantity. There, the vertical axis is the minimum discrete Fisher information over the period window used in the numerical scan. For the qubit range shown in Fig. 2(b), that window extends up to $r_{\max}(n) = n^2$, so the relevant QFT comparator is the worst case within the same scan rather than a single fixed value of r .

The leading term in Theorem 3 is monotone in R , and in the sampled regime $R \approx 2^n/r$. Consequently the smallest QFT Fisher information inside the scan window is attained near the largest sampled period, which yields the window-matched estimate

$$I_{\text{QFT,win}}(n) \approx \frac{4\pi^2}{9} \left(\left\lfloor \frac{2^n}{n^2} \right\rfloor^2 - 1 \right). \quad (\text{E1})$$

If one keeps only the dominant exponential factor, this becomes

$$\ln I_{\text{QFT,win}}(n) = (2 \ln 2) n - 4 \ln n + O(1), \quad (\text{E2})$$

so the asymptotic QFT exponent is $k_{\text{QFT},\infty} = 2 \ln 2 \approx 1.386$, while a direct log-linear fit over the finite window $n = 7, \dots, 14$ gives the effective slope $k_{\text{QFT,win}} = 1.067$.

This observation suggests a simple diagnostic for the numerical data in Fig. 2(b). If the plotted curves retain a stable exponential dependence on the qubit count over the sampled window, then the transformed data should be well approximated by

$$\ln I_{\min}(n) = kn + b. \quad (\text{E3})$$

We therefore perform ordinary least-squares fits to the log-transformed mean values of each curve shown in Fig. 2(b). Figure 4 displays the resulting regressions.

Circuit	qubits	k	95% CI	R^2
ph1	7–14	0.378	[0.306, 0.449]	0.965
optimized 1ph	7–12	0.443	[0.253, 0.632]	0.913
ph1_random	7–14	0.386	[0.318, 0.453]	0.970
ph_random	7–14	0.406	[0.361, 0.451]	0.988
ph_random_phase	7–14	0.363	[0.319, 0.407]	0.986

TABLE I. Effective exponents obtained from the log-linear model $\ln I_{\min}(n) = kn + b$ applied to the numerical Fisher-information curves in Fig. 2(b).

The fitted lines account for nearly all of the log-scale variance across the sampled window. The coefficient of determination ranges from 0.913 to 0.988, the log-space root-mean-square residual lies between 0.101 and 0.233, and every fitted slope is significantly positive with $p \leq 2.9 \times 10^{-3}$. In that sense, a linear regression on $\ln I_{\min}$ is justified as a finite-window descriptive model for these curves. At the same time, the fitted quantity should be interpreted as an effective exponent on the restricted range $n = 7, \dots, 14$, not as a claim about the ultimate asymptotic law.

The comparison with the QFT benchmark is unambiguous. Even after matching the period window used in the numerical scan, the QFT reference fit remains much steeper, with $k_{\text{QFT,win}} = 1.067$ and $R^2 = 0.9966$. The empirical circuit families therefore grow substantially more slowly: their fitted exponents occupy only 34%–41% of the window-matched QFT slope, and only 26%–32% of the asymptotic QFT exponent $2 \ln 2$. Among the plotted families, optimized 1ph has the largest point estimate $k = 0.443$, although its confidence interval is widest because that dataset currently stops at 12 qubits. Among the curves available through 14 qubits, ph_random has the largest fitted exponent, $k = 0.406$.

This comparison sharpens the interpretation of Fig. 2(b). The HP-family circuits retain a clear exponential-in- n growth trend in the minimum Fisher information over the sampled window, which is why the log-linear fit works well, but they do not reproduce the QFT growth rate. Their advantage in the present setting is therefore not that they match the full QFT benchmark asymptotically, but that they retain a nontrivial and practically usable fraction of that information while remaining much shallower.

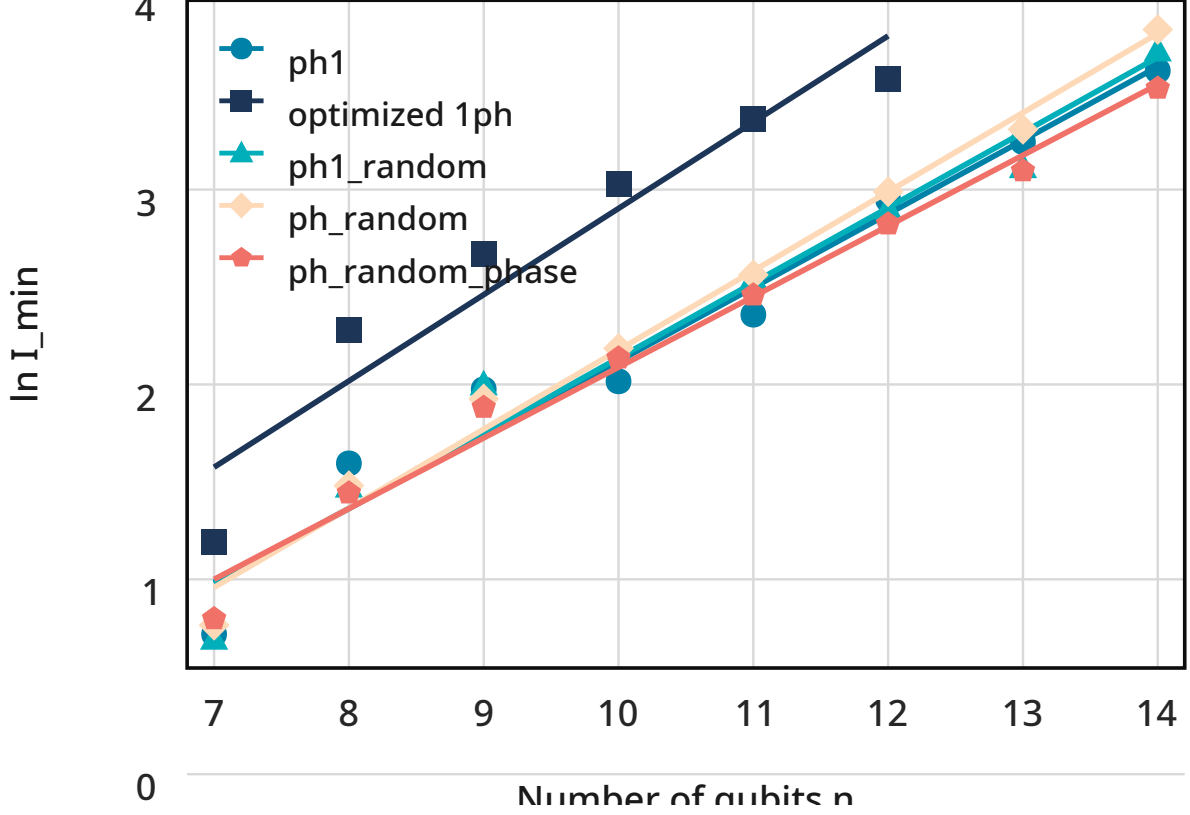


FIG. 4. Log-linear fits of the qubit-scaling curves in Fig. 2(b). Markers denote the mean Fisher-information values plotted in Fig. 2(b), after applying the transformation $I_{\min} \mapsto \ln I_{\min}$, and solid lines denote the corresponding least-squares fits $\ln I_{\min}(n) = kn + b$.

F. Noise-robustness experiment details

We evaluate noise robustness using the previously trained 9- and 10-qubit HP-1 plus neural-decoder pipelines from the period-recovery study. For each system size, we use a measurement budget of $1024n^2$ samples per instance, giving 82944 measurements for $n = 9$ and 102400 for $n = 10$. The candidate periods are $r \in \{9, \dots, 80\}$ for 9 qubits and $r \in \{10, \dots, 99\}$ for 10 qubits. Probability distributions are generated from the exact subgroup support of the trained HP-1 circuit, and the neural decoder checkpoint is selected using noiseless validation accuracy. The selected checkpoints occur at epoch 3 with validation top-1 accuracy 0.9959 for 9 qubits, and at epoch 10 with validation top-1 accuracy 1.0000 for 10 qubits.

The evaluation set holds out 3 shifts for each candidate period and generates 8 independent noisy redraws for each held-out shift. This yields $72 \times 3 \times 8 = 1728$ test examples for 9 qubits and $90 \times 3 \times 8 = 2160$ test examples for 10 qubits. To probe robustness after the shallow quantum transformation, we insert a single global depolarizing channel after the full HP-1 circuit and before measurement. At the probability level, this is implemented as

$$\tilde{p}_\eta(x|r, s) = (1 - \eta)p(x|r, s) + \eta \frac{1}{2^n}, \quad (\text{F1})$$

where $p(x|r, s)$ is the ideal output distribution for period r and shift s , and η is the depolarizing strength. For each noisy distribution $\tilde{p}_\eta$, we resample $1024n^2$ outcomes and feed the resulting bit-string batch into the fixed neural decoder. The sweep uses the values

$$\eta \in \{1, 0.464, 0.215, 0.1, 0.0464, 0.0215, 0.01, 0.00464, 0.00215, 0.001\}. \quad (\text{F2})$$

Several features of Table II are worth noting. First, performance is essentially unchanged throughout the weak-to-moderate noise regime and remains near the noiseless baseline up to $\eta \approx 0.215$. Second, substantial degradation

η	9-qubit accuracy	10-qubit accuracy
1.000	0.0139	0.0111
0.464	0.4913	0.8810
0.215	0.9948	1.0000
0.100	0.9954	1.0000
0.0464	0.9959	1.0000
0.0215	0.9942	1.0000
0.0100	0.9954	1.0000
0.00464	0.9959	1.0000
0.00215	0.9948	1.0000
0.00100	0.9948	1.0000

TABLE II. Held-out period-recovery accuracy under post-HP-1 global depolarizing noise.

appears only once the noisy distribution is strongly mixed with the uniform distribution: at $\eta = 0.464$, the accuracy drops to 0.4913 for 9 qubits and 0.8810 for 10 qubits. Finally, when $\eta = 1$, the output distribution becomes exactly uniform and is therefore independent of both the period and the shift. In that limit, all subgroup information is erased, so the best possible behavior is chance-level guessing over the candidate periods. The observed accuracies 0.0139 and 0.0111 agree exactly with the corresponding chance levels $1/72$ and $1/90$, confirming that the numerical results are consistent with the depolarizing-noise model.